

Shaheer Tahir Rao

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PROFESSIONAL SUMMARY

Electrical Engineering graduate specializing in power systems and industrial energy infrastructure. Built hands-on substation, switchgear, and SCADA experience through internships at NESPAK and Pak Elektron Limited, backed by a strong foundation in power network analysis, protection concepts, and electrical design documentation.

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

September 2022 – May 2026

BS Electrical Engineering

PROFESSIONAL EXPERIENCE

National Engineering Services Pakistan (PVT) Limited

July 2025 – August 2025

Technical Intern - SCADA and Telecom Division

- Reviewed SCADA/telecom design submittals for WAPDA hydropower projects (Dasu 4320 MW; Diamer-Bhasha 4500 MW) and produced technical review summaries
- Verified ELV/ICT AutoCAD drawings and prepared coordination redlines for layouts and routing

Pak Elektron Limited

July 2024 – August 2024

QA / QC Intern - Switchgear Division

- Audited schematics and performed basic parameter checks for CT/VT, relays, interlocks, and earthing
- Supported switchgear documentation and drafting, including verifying basic panel ratings (voltage/current) from nameplates and drawings
- Submitted 2 AutoCAD deliverables: Star-Delta motor starter/protection schematic and a switchgear technical drawing

ACADEMIC PROJECTS

Automated CPR Machine – (Final Year Project)

September 2025 – May 2026

- Developing an automated CPR machine benchmarked to LUCAS 3 (Stryker), targeting 102/111/120 ± 2 cpm, 45–53 ± 2 mm depth, and 50 $\pm 5\%$ duty cycle
- Implementing a dual-actuation setup: DC geared motor for compression torque/force control and a stepper motor for adjustable chest-height/positioning
- Integrating real-time vitals monitoring (e.g., heart rate and blood pressure) via onboard sensors for live display/logging during operation

Linear Variable Differential Transformer

September 2024 – January 2025

- Designed and wound a mini LVDT (1 primary + 2 matched secondaries in series- differential connection) using 28-AWG copper with ~ 120 turns per coil
- Confirmed correct differential behavior with a moving solenoid/core: ± 45.9 mA output on both sides with polarity reversal across the null

Microphone Amplifier

January 2024 – May 2024

- Designed and built a 2-stage BJT microphone amplifier, using RC coupling (coupling capacitor) between stages for signal amplification
- Implemented a push-pull BJT output stage to drive a 4 W, 40 Ω speaker (≈ 12.65 V_{rms}, 0.316 A_{rms} at full power)

Spot Welder

January 2024 – May 2024

- Developed a DIY resistance spot welder using the NY-D01 digital controller and implemented adjustable weld pulse timing: 01–50 cycles (20 ms/cycle) with foot-switch triggering
- Enabled current/intensity adjustment: 30–99% (controller type 40 A) and tuned settings through iterative test welds
- Produced project deliverables: wiring/schematic and enclosure integration for repeatable operation

COMMUNITY SERVICE

Government Boys High School - (Serhali Khurd, Kasur)

Volunteer Academic Instructor

June 2023 – July 2024

- Volunteered as an academic instructor for secondary and higher secondary students (Classes 9–12) in a rural government school
- Delivered structured instruction in Mathematics and English aligned with the Board of Intermediate and Secondary Education (BISE) curriculum

TECHNICAL SKILLS

Programming Languages: C++, Python, Matlab, Verilog

Softwares: Proteus, LTspice, Multisim, PLECS, TeXstudio, Simulink, AutoCAD, COMSOL, LVDAC-EMS, ETAP, Power World Simulator, TIA Portal, Canva, MS Office